

OPENAI × NAMASTEDEV HACKATHON · JULY 2026

# Myriox

Cast personas, not particles.

Autonomous multi-agent simulation for testing physical spaces before they're built.

Built with OpenAI Codex

## THE PROBLEM

# Accessibility failures get discovered after construction

- **Too late, too expensive.** Flow and accessibility problems in libraries, clinics, stores, and offices are almost always found after the building exists — when fixing them means demolition and retrofits.
- **Particles aren't people.** Traditional crowd-simulation tools model everyone the same way. They find narrow corridors. They don't notice that a wheelchair user is permanently stuck at a stairwell with no ramp.
- **Human audits are slow.** Thorough, but they happen far too late in the design process to change anything.

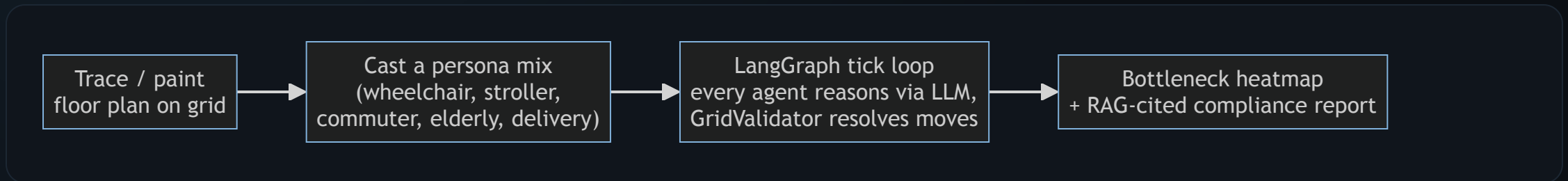
## THE PRODUCT

# Give every persona a real LLM brain

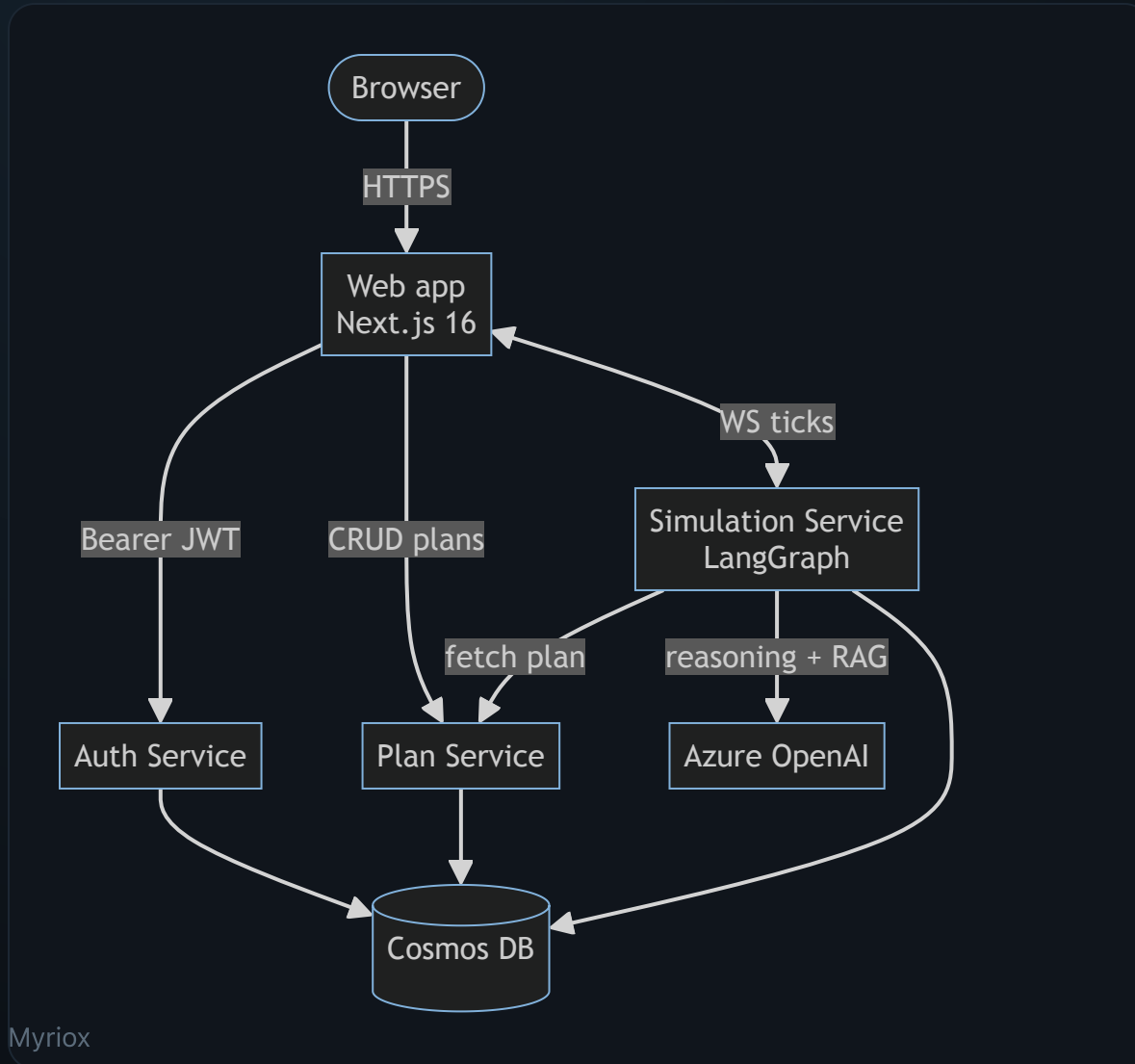
- Trace or paint a floor plan on a grid.
- Cast a mix of AI personas — **wheelchair user, parent with stroller, rushed commuter, elderly visitor, delivery worker** — each with its own mobility profile and objective.
- Every tick, every agent reasons about its surroundings via a real LLM call — not a physics rule.
- Get back a **bottleneck heatmap** and a **compliance report** citing the exact ADA/IFC clause behind each genuine finding.

## HOW IT WORKS

# Trace → Cast → Simulate → Audit



# Three microservices, one shared brain



- **Frontend:** Next.js 16, Tailwind, Framer Motion.
- **Backend:** independent FastAPI services on Azure Container Apps.
- **Orchestration:** LangGraph — agents reason concurrently, a validator resolves the tick.
- **Data:** Cosmos DB.
- **AI:** Azure OpenAI chat + embeddings.

## PROOF

# A real 1885 library floor plan, a real cited finding

- Traced from a genuine Library of Congress floor plan (public domain): the back corridor and stairwell, with no ramp.
- Actual run: 2 agents, 36 ticks, ~4 minutes end to end. The wheelchair-mobility agent never got past the stairwell.
- Result: **5 findings, all citing ADA 4.13.9.**

"A wheelchair user was blocked with only stairs as the apparent route forward; no accessible alternative route was available at this location."

WHAT'S NEXT

# Roadmap

- **01** Blob storage + CDN for reference images (currently inline data URIs).
- **02** Streamed partial-tick results for large, responsive runs.
- **03** More jurisdictions in the compliance clause database (IFC, regional codes).
- **04** Multi-seat org sharing on top of existing org-scoped auth.

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